Data Analysis with Python

Course Introduction & Syllabus

Instructor: Eric Li | Semester: Fall 2026

Wechat Group!

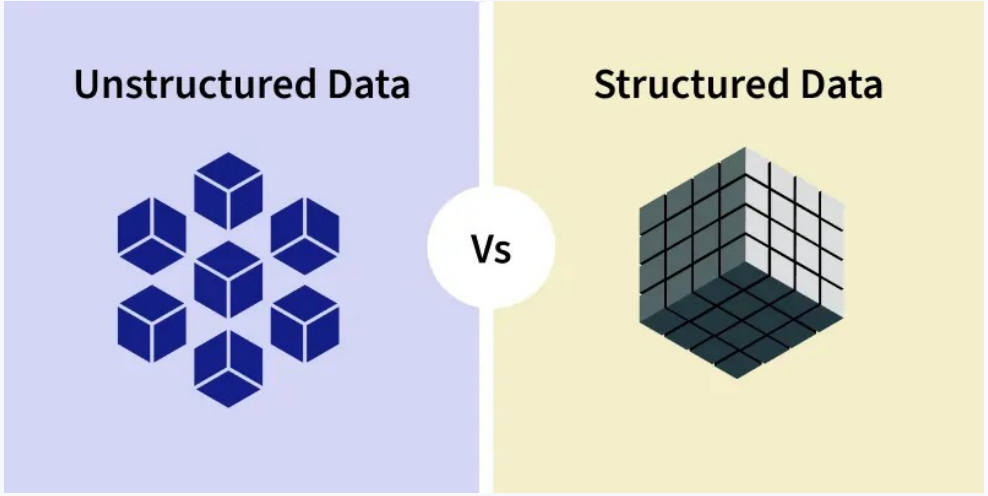
1. Understanding Data Analysis

What is Data Analysis?

The process of **cleaning, transforming, modeling, and inspecting** data to discover useful insights.

What kinds of data?

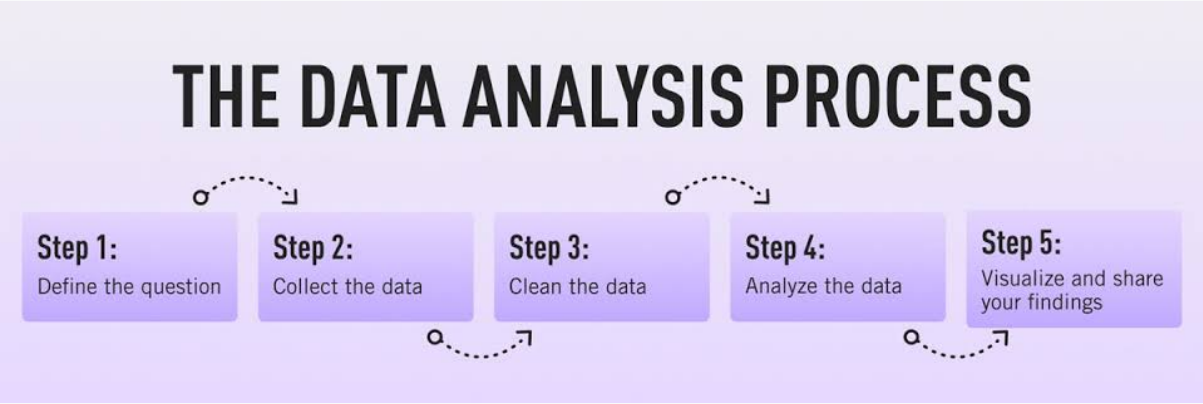
We are concerned about **structured data**, which is a deliberately vague term that encompasses tabular or spreadsheet-like data, commonly stored in relational databases or text files. **Unstructured data**, on the other hand, still exist widely and present a great challenge to data analysis.



1. Understanding Data Analysis

The Core Workflow: This might not be a linear process!

- **Question**: Defining questions you are curious about.
- **Collection**: Gathering raw metrics, logs, or text files.
- **Cleaning**: Resolving missing values, structural errors, and anomalies (70% of the job!).
- **Analysis**: Performing EDA to map trends, distributions, and correlations.
- **Communication**: Crafting visual charts to drive strategic decisions.



2. Why Python for Data?

Why Python?

- **Highly Readable**: Clean syntax that looks like English, keeping focus on logic over syntax errors.
- **Massive Ecosystem**: Thousands of packages optimized for math, statistics, and machine learning.
- **Industry Standard**: The dominant language globally for Data Science and modern business analytics.
- **Seamless Integration**: Connects directly with SQL databases, web frameworks, and cloud workflows.

Why not Python?

- Python code, as it is an interpreted language, runs substantially slower that code written in a compiled language like Java or C++.
- Python is not suitable for building highly concurrent, multi-threaded applications due to the GIL (global interpreter lock) preventing it from executing multiple instructions (threads) at a time.

3. The Ecosystem Stack

The Core Tools We Will Use:

- **Python**: The keystone programming language.
- **NumPy**: Powerhouse array computing to handle underlying mathematical vectors efficiently.
- **pandas**: The 'Excel of Python'—essential for tabular data frames, sorting, and cleaning.
- **matplotlib & seaborn**: Tools for engineering production-grade visual narratives and graphs.
- **statsmodels & scikit-learn**: Modeling toolkits for uncovering deeper relationships and feature engineering.
- **Jupyter Notebook**: Our workspace—an interactive ecosystem combining code, documentation, and charts.

4. Course Agenda & Scope

Fall 2026 Semester Timeline:

- Weeks **1–3** (Sep 1 - Sep 18)

  U0 Course Intro & Syllabus, U1 Installation & Setup, U2 Python Language Basics

- Weeks **4–7** (Sep 21 - Oct 16)

  U3 NumPy Basics, U4 pandas Basics

- Weeks **8–10** (Oct 19 - Nov 6)

  U5 Data Loading and Cleaning, Midterm

- Weeks **11–13** (Nov 9 - Nov 27)

  U6 Data Wrangling, U7 Data Aggregation

4. Course Agenda & Scope

Fall 2026 Semester Timeline [cont.]:

- Weeks **14–17** (Nov 30 - Dec 25)

  U8 Plotting, U9 Time Series, U10 Modeling Libraries

- Weeks **18–20** (Dec 28 - Jan 15)

  U11 Data Analysis Examples, Final

Spring 2027 Semester Anticipation:

- Machine learning with Python (?)

5. Grading Structure

Class Assignments

You will have 2 assignments in this class:

- A1: Reading reflection, due **Sep 11**, worth 10%
- A2: Data analysis project, due **Dec 31**, worth 90%

Exams

Midterm and final will be usual on-paper exams. They will not only test your **language proficiency** of Python, NumPy, pandas, and others, but also your understanding of the data analysis processes as well as mastery of data analysis techniques in **answering practical questions**.

- Exams will be composed of MCQs and short response questions.

5. Grading Structure

Sample Question:

Which of the following is not part of Data Analysis?

- A) Building statistical models and data visualizations
- B) Picking a desired conclusion for the analysis
- C) Fixing incorrect values and removing invalid data
- D) Transforming data into an appropriate data structure

It is more than a programming class…

6. AI Framework Policy

AI as a Copilot:

- **Embrace the Tooling**: Generative tools are excellent assistants for fixing syntax and explanation.
- **The Owner Rule**: You can utilize AI, but you must comprehend every statement submitted.

  If selected at random, you must cleanly explain your logic step by step.
- **System Limits**: AI frequently invents patterns or outputs over-engineered syntax. Verify everything.
- **Absolute Violations**: Ghostwriting assignments without core engagement breaks integrity protocols.

7. Assignment 1 - Reading Reflection

Assignment requirements:

Read the following three articles, and write a reading reflection of **no less than 200 words**. Your response may endorse, criticize, question, or expand on the authors' opinions, and needs to show your personal thoughts and feelings towards the role of AI in data analysis. Be bold and creative!

- Data Analysis in the Age of AI, by Yusuf Saber
- Don't Let AI Make Bad Analytics Worse, by Steven Tadelis
- Why Data Analysts Remain Relevant in the Age of AI, by Malcolm Hawker

Grading:

This assignment is due on **Sep 11 12:59pm**. To submit, email your response (can be in any format: word/pdf/text box…) to zhengyi.li@ssbs.sh.cn, and be sure to include your name and class.

This assignment is worth **10 pts**, and will be graded on your level of completion and how much thought you've put into this topic. Late turn-in will receive a deduction of 1 pt per day until a 4 pts maximum (unless excused beforehand).

Any question?